

Day 3-4 Teaching Deck — Exploratory Data Analysis (EDA) & Model Training

This is a **combined Day 3 and Day 4, content-first teaching presentation**.

Theoretical concepts are kept **concise and early**, followed by **hands-on application** using the **Diamond Price Prediction** Kaggle notebook.

The focus is to help senior engineers: - Understand *why* EDA exists - Learn **practical EDA techniques used in industry** - Apply EDA insights to **improve a real ML model**

Slide 1 — Title

Day 3-4: Exploratory Data Analysis (EDA) & Model Training

Understanding data before building models

Slide 2 — Quick Recap (Day 1 & Day 2)

From earlier sessions: - AI systems learn patterns from data - Machine Learning success depends heavily on data quality - Supervised learning uses labeled data - Regression predicts numeric values

Key takeaway:

Before choosing a model, we must understand the data deeply.

Slide 3 — Why EDA Is Critical

Exploratory Data Analysis (EDA) helps us: - Understand the structure of data - Detect errors and anomalies - Identify relationships between variables - Decide what features matter

Reality check:

Most ML performance gains come from better data understanding, not better algorithms.

Slide 4 — What Is EDA?

EDA is the process of: - Summarizing data statistically - Visualizing distributions and relationships - Asking questions about the data

EDA happens **before** and **during** model building.

Slide 5 — Types of EDA Techniques

EDA techniques broadly fall into: - **Univariate analysis** (single variable) - **Bivariate analysis** (two variables) - **Multivariate analysis** (many variables)

We explore: - Distributions - Relationships - Outliers - Missing values

Slide 6 — Univariate Analysis

Focuses on a single feature: - Mean, median, variance - Distribution shape - Outliers

Examples: - Price distribution - Carat distribution

Purpose: Understand scale and spread.

Slide 7 — Bivariate Analysis

Focuses on relationships between two variables: - Feature vs target - Feature vs feature

Examples: - Carat vs Price - Cut vs Price

Purpose: Identify predictive relationships.

Slide 8 — Multivariate Analysis

Looks at multiple variables together: - Correlation matrices - Pair plots - Feature interactions

Purpose: Understand combined effects and redundancy.

Slide 9 — EDA Techniques You Should Know

Common industry techniques: - Histograms and KDE plots - Box plots - Scatter plots - Correlation heatmaps - Group-by aggregations

These techniques guide feature engineering.

Slide 10 — Recommended EDA Resources (Mandatory Reading)

Participants must review the following:

1 Kaggle Discussion: *Exploratory Data Analysis*
<https://www.kaggle.com/discussions/general/231317>

2 Interactive EDA Notebook
<https://www.kaggle.com/code/dcstang/interactive-exploratory-data-analysis-eda>

3 Advanced EDA for Tabular Data
<https://www.kaggle.com/code/vbmokin/eda-for-tabular-data-advanced-techniques>

Slide 11 — What You Should Extract From These Links

From the EDA resources, focus on: - How distributions are analyzed - How categorical variables are explored
- How correlations guide feature selection - How EDA insights influence modeling decisions

Do not copy plots blindly — understand *why* they are used.

Slide 12 — Transition: From EDA to Modeling

EDA answers questions like: - Which features matter most? - Which features are noisy or redundant? - Are transformations needed?

These answers directly affect model performance.

Slide 13 — Hands-On Dataset: Diamond Price Prediction

Base notebook: <https://www.kaggle.com/code/karnikakapoor/diamond-price-prediction>

Task: - Use this notebook as a **starting point only** - Do additional EDA beyond what already exists

Slide 14 — What You Must Add (EDA Improvements)

Apply **at least 3–4 new EDA techniques**, such as: - Deeper correlation analysis - Better categorical feature exploration - Outlier detection and reasoning - Feature transformations (log, scaling)

Your EDA should inform modeling decisions.

Slide 15 — Model Training Expectations

After EDA: - Train a regression model - Clearly explain: - Why you chose the model - What EDA influenced this choice

Model complexity is less important than reasoning.

Slide 16 — What We Are Evaluating

We are evaluating: - Quality of EDA reasoning - Ability to explain insights - Connection between EDA and model decisions - Clean, readable notebooks

Not just model accuracy.

Slide 17 — Common Mistakes to Avoid

- Jumping straight to modeling
- Blindly copying EDA plots
- Ignoring categorical variables
- Treating EDA as optional

EDA is part of engineering, not decoration.

Slide 18 — Day 4 Focus: Refinement & Reflection

On Day 4 you should: - Refine EDA based on insights - Improve feature handling - Retrain the model - Compare results before vs after EDA

Slide 19 — Deliverables

Each participant must submit: - A Kaggle notebook or GitHub repo - Clear EDA section - Model training section - Short explanation of decisions

Slide 20 — Reflection Questions

Be ready to answer: - What EDA insight mattered most? - What surprised you in the data? - What would you do with more data?

Slide 21 — Key Takeaways

- EDA is foundational, not optional
- Good EDA leads to better models
- Most ML value comes from understanding data
- Clear explanations matter more than fancy models

End of Day 3–4 Teaching Presentation